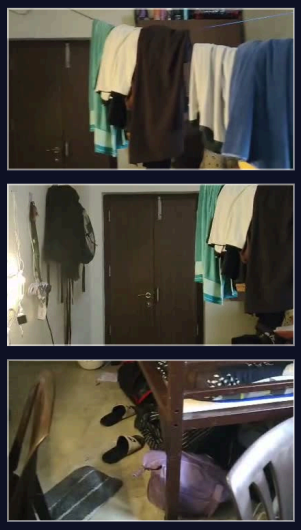
Room3D

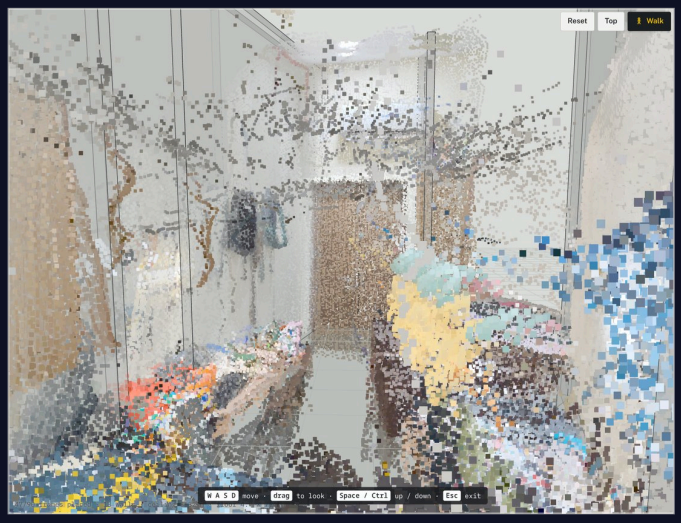
One handheld phone video of a room in. A measurable 3D model you can walk through in a browser out.

INPUT



CPU

OUTPUT



Frames from one handheld phone clip (WhatsApp, 1.28 AM)

The reconstructed room in the browser viewer (free-roam)

1
handheld phone video is the only input

0
depth sensors, LiDAR, targets or manual measurements

90
sharp frames at most are used per clip

CPU
only, from video to finished model

01 Overview and scope

Room3D turns a single continuous video from an ordinary phone rear camera into three outputs, all in real-world metres: a dense colored point cloud of the room; the floor, ceiling and vertical walls as a clean axis-aligned floor plan; and furniture as labeled, oriented 3D bounding boxes with measured width × depth × height.

A compact browser viewer renders these with per-layer toggles and a first-person free-roam mode, so the model is measurable and explorable, not merely viewable. The system covers the full chain: a capture procedure (Section 2), a seven-stage CPU reconstruction pipeline (Section 4), an explicit scene format (Section 3), and the viewer (Section 7).

KEY CONTRIBUTIONS

Metric from monocular video
Per-frame mono-depth is bound to the Structure-from-Motion scale, so output is in real metres, not an arbitrary unit.

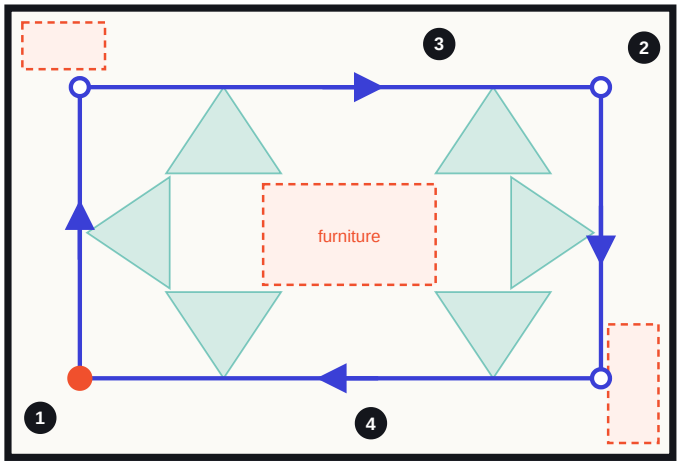
Flying-pixel filter
Removes the dominant monocular-depth noise before fusion, at almost no cost.

Manhattan floor plan
Straight, right-angled walls recovered from tall-cell evidence in the plan view.

Multi-view furniture boxes
Partial masks from many frames fused into complete, measured, oriented boxes.

02 Standard Operating Procedure (capture)

Reconstruction quality is set at capture time. The SOP is a short, repeatable procedure designed to steer the operator away from the failure modes of Structure-from-Motion and monocular depth. Follow the path in Figure 1 and the checklist below.



1 Start and end, same spot
Returning closes the loop and keeps the plan square.

2 Pause at every corner
Each corner is then seen from two directions.

3 Slow loop, camera inward
About 0.3–0.5 m/s, consecutive frames overlap.

4 Keep wall base in view
Floor-to-wall boundary of every wall, at some point.

SIDE VIEW

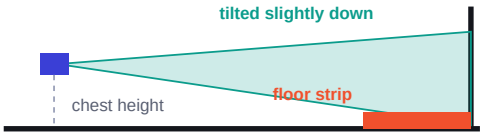


FIGURE 1 Capture path. A single slow loop around the perimeter, camera facing inward (teal wedges are the field of view), pausing at corners and returning to the start.

DO

Camera. Any phone rear camera, landscape, 1080p at 30 fps. Higher is fine, the pipeline down-samples. Let auto-exposure and focus settle first.

Height. About chest level, tilted slightly down so a strip of floor is visible in every frame.

Duration. 20–60 s is enough; up to 90 sharp frames are sampled.

Lighting. Even, diffuse, room lights on. Avoid strong window backlight.

AVOID

Fast motion. Whip-pans and jumps break sequential matching and blur frames.

Blank walls. Close-ups of textureless surfaces give nothing to track.

Mirrors and glass. They corrupt depth and feature matching.

Moving subjects. People or pets in frame.

PARAMETER	VALUE	WHY IT MATTERS
Speed	0.3–0.5 m/s	Frames are matched sequentially, so neighbours must share most of their view.
Resolution	1080p, 30 fps	Enough detail for SIFT features; frames are resized to 900 px anyway.
Clip length	20–60 s	Covers the full perimeter at the target speed; the frame budget caps at 90.
Loop closure	Return to start	Keeps the recovered plan square instead of drifting open.
Corners	Brief pause	Each corner observed from two directions improves wall fitting.

03 System architecture

Three parts, shown in Figure 2: (a) an offline reconstruction pipeline, running on CPU, that turns the video into geometry; (b) a compact, explicit scene representation; and (c) a browser viewer that renders and roams it. All heavy computation stays in part (a); the viewer only reads the compact scene.

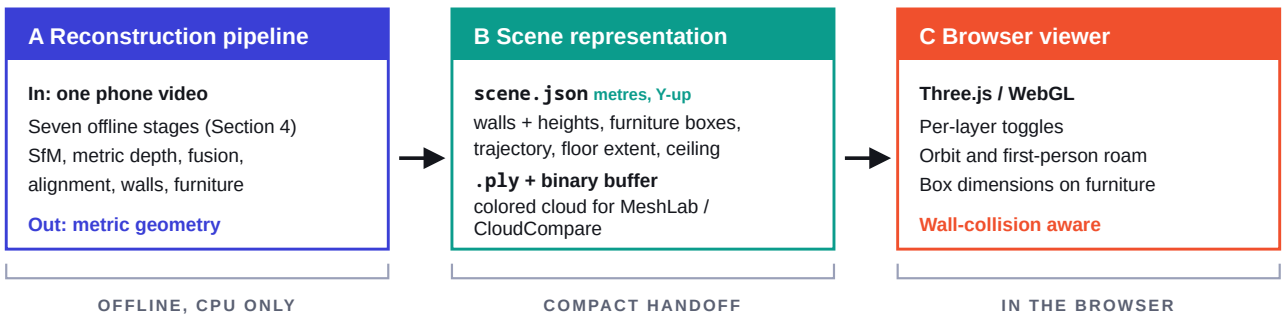


FIGURE 2 System architecture. The scene representation is the only interface between the offline pipeline and the viewer.

The representation is deliberately explicit. A single `scene.json` holds the wall segments with heights, the furniture boxes (centre, size, yaw, label), the camera trajectory, the floor extent, and the ceiling height, all in metres in a Y-up frame with the floor at y = 0. The colored point cloud is stored as a standard `.ply` that opens directly in MeshLab or CloudCompare, and as a compact binary buffer for fast loading in the viewer.

04 Reconstruction pipeline

Seven CPU stages, each consuming the previous stage's output (Figure 3). The two highlighted rows are where the method gets its metric scale and its clean geometry.

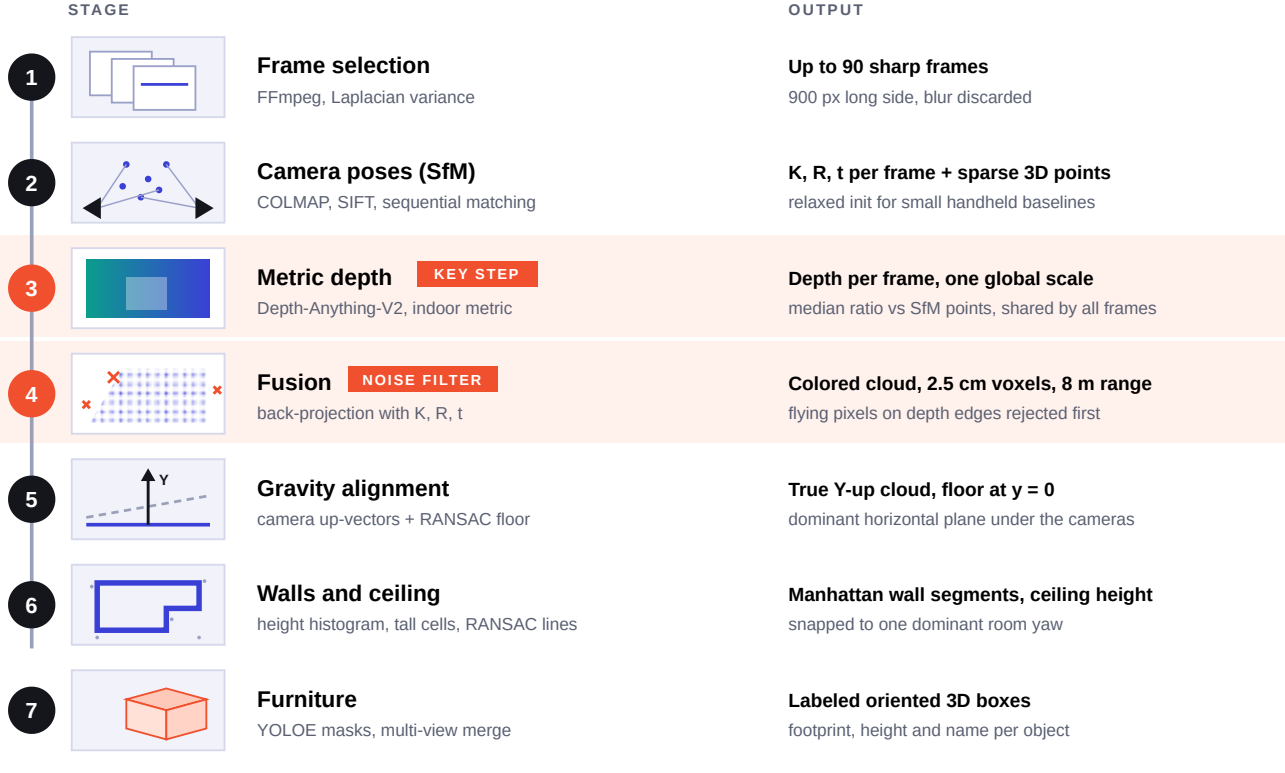


FIGURE 3 The seven-stage pipeline. Left: stage, with a sketch of its data. Right: its output. Red rows mark the metric-scale binding and the noise filter.

04 Stage details

1 Frame selection

FFmpeg samples at an adaptive rate so any clip yields up to 90 frames. Within each small candidate group the sharpest frame is kept (highest Laplacian variance), which discards motion blur. Frames are resized to a 900 px long side.

2 Camera poses (SfM)

COLMAP extracts SIFT features under one shared radial-distortion camera model, matches them sequentially across neighbouring frames, and maps incrementally with deliberately relaxed initialisation, since handheld baselines are small. Output: per-frame K, R, t and a sparse cloud with 2D-to-3D correspondences.

3 Metric depth: the step that makes it metric

Depth-Anything-V2 (indoor-metric variant) predicts a dense depth map per frame. Monocular depth is only correct up to an unknown per-frame scale, so each map is aligned to the SfM points by the median ratio between predicted and SfM depth at the reprojected keypoints. A single global scale is then shared across all frames. This is what makes the output genuinely metric, and it keeps depth and poses mutually consistent instead of drifting frame to frame.

4 Fusion

Each depth map is back-projected into world space with K, R, t. A flying-pixel filter rejects pixels on steep depth discontinuities (high relative gradient), the stray points that monocular depth smears along the view ray and the dominant noise source. Points beyond 8 m are dropped, and the union is voxel-down-sampled at 2.5 cm.

5 Gravity alignment

The up direction is estimated from the cameras' up-vectors and refined by RANSAC-fitting the dominant horizontal plane beneath the cameras, the floor. The cloud is rotated into a true Y-up frame and translated so the floor lies at y = 0.

6 Walls and ceiling

The ceiling height is read from the upper peak of the height histogram. The floor plan is binned into cells, and any cell whose points span a tall vertical range counts as wall evidence. The room's dominant yaw is estimated from the gradients of that floor-plan image, and wall segments are fitted as RANSAC lines snapped to the Manhattan directions, since indoor rooms are mostly axis-aligned. Noisy evidence becomes a clean, straight floor plan.

7 Furniture

A prompt-free open-vocabulary segmenter, YOLOE, masks furniture in every frame. Masks are cleaned by erosion and a depth-median gate that removes background bleed, then back-projected to 3D and lifted into the aligned frame. Observations of one object across frames are merged by label and proximity, and each merged object is fitted with an oriented bounding box (minimum-area rectangle in plan, snapped to the room yaw). Synonymous detector labels collapse to a canonical vocabulary, so one real object becomes exactly one box with its measured footprint, height and name.

05 Technical approach and design decisions

Monocular depth over multi-view stereo
Dense MVS is slow on CPU and fails on the textureless walls common indoors. A learned metric-depth prior fills them densely.

Global scale sharing
Per-frame scale drifts. One median scale keeps the floor flat and the cloud coherent.

Flying-pixel rejection
Removes the largest monocular-fusion noise source cheaply, before fusion rather than after.

Manhattan floor plan
Snapping to one dominant yaw yields straight, right-angled walls.

Multi-view box fusion
Any single mask is partial. Merging across frames gives complete, trustworthy boxes.

CPU-only throughout
Portable and reproducible on commodity hardware.

06 Technology stack

Python 3
Implementation language
python.org

COLMAP (via pycolmap)
Structure-from-Motion camera poses
colmap.github.io

Depth-Anything-V2
Monocular metric depth, indoor variant
github.com/DepthAnything

YOLOE (Ultralytics)
Prompt-free open-vocabulary segmentation
docs.ultralytics.com/models/yoloe

PyTorch
Runs the neural models on CPU
pytorch.org

OpenCV, NumPy, SciPy
Geometry and image processing
opencv.org

FFmpeg
Frame extraction from video
ffmpeg.org

Three.js (WebGL)
In-browser 3D viewer
threejs.org

07 Web viewer and free-roaming

The viewer renders the compact scene in the browser with Three.js and WebGL. Each layer toggles independently, and furniture boxes show their measured dimensions.

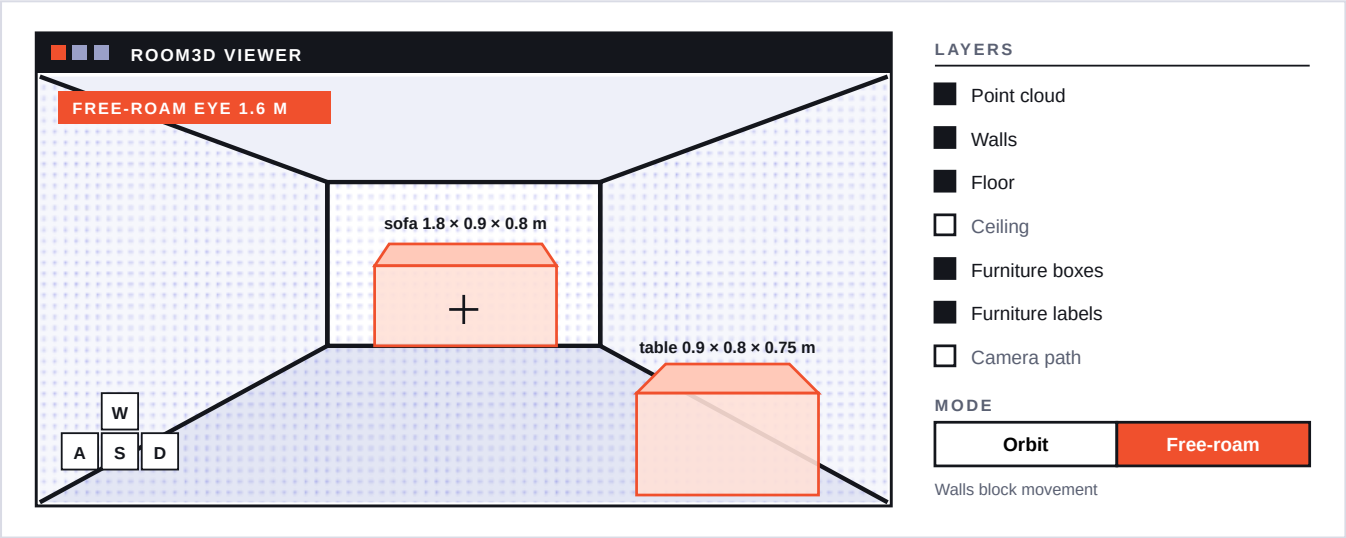


FIGURE 4 Viewer in free-roam mode (schematic): measured furniture boxes inside the colored cloud, layer toggles on the right, and wall-limited first-person movement.

Point cloud **Walls** **Floor** Ceiling **Furniture boxes** **Furniture labels** Camera path

Orbit

Whole-room overview. Double-click recentres the view, and a top-down floor-plan view is available.

Free-roam (first person)

Enters at eye height (1.6 m). W A S D or arrow keys move, mouse-drag looks around, Space and Ctrl change height. Movement is blocked by the reconstructed walls and kept inside the room, so a reviewer can walk through the space.

FIDELITY AND LIMITATIONS

Monocular depth gives centimetre-level accuracy, not survey-grade accuracy. Reflective and glass surfaces, very dark rooms, and large blank walls degrade the result. The SOP in Section 2 is designed to steer capture away from exactly these failure modes.